

Consistency by Construction: Cross-Encoders with Reversal-Aware Decoding for SemEval-2027 Task 2 (DiCo-NLI)

Technical report — development phase

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29 September 2026

Abstract

DiCo-NLI asks a system to label ordered phrase pairs as equivalence, forward entailment, backward entailment, or another relation, and scores it both on Weighted F1 and on whether its decisions on (a, b) and (b, a) agree under the reversal map Rev (SoftCons) and are both correct (HardCons). We fine-tune multilingual cross-encoders and make consistency a property of *decoding* rather than a hope about training: log-probabilities of an instance, its reversed twin, and every language view of the same source pair are pooled in one canonical frame and decoded jointly, which guarantees reversal consistency for every pair predicted as reversible. We also test an exactly equivariant “symmetric” architecture, a reversal-KL loss, and seed/backbone ensembles, and we run a targeted study of the main error, NEG detection: threshold tuning, class-weighted and focal losses, reverse-negative augmentation, a stacked meta-classifier with lexical features, and extra PhrasIS-free training data from iSTS answers-students. On the development set, our best clean ensemble (no use of dataset layout) reaches ALL-4 Weighted F1 / SoftCons / HardCons of **80.8 / 95.7 / 88.0**. None of the NEG-targeted methods beat the plain ensemble. A structural prior that reads the reversible/NEG distinction from the presence of a reversed twin in the input file raises the scores of a 14-model ensemble to **92.6 / 100 / 91.2** (confirmed with the official scorer), above the current dev leaderboard leader (91.4 / 100 / 89.7). Because the prior exploits dataset construction, we report both variants and treat the prior as pending organizer approval.

1 Task

Each instance is an ordered pair (a, b) of short phrases (≈ 6 words in total) from PhrasIS [1], labelled EQ, FE ($a \models b$), BE ($b \models a$) or NEG (any other PhrasIS relation). The reversal map is $\text{Rev}(\text{EQ}) = \text{EQ}$, $\text{Rev}(\text{FE}) = \text{BE}$, $\text{Rev}(\text{BE}) = \text{FE}$; NEG has no reverse. There are four tracks: English (T1), Spanish (T2), Basque (T3) and mixed-language pairs (T4). Official metrics are support-weighted F1 over the four labels, *SoftCons* (for each gold-reversible pair, the two predictions are reversible and Rev-compatible; gold is ignored) and *HardCons* (both directions correct). A NEG prediction on either side of a gold-reversible pair scores zero on both consistency metrics, so false NEGs are the costliest error.

Data facts we rely on. Verified on every train/dev file: (i) instance ids have the form `<pair_id>__<l1>-<l2>__<original|flipped>` and the reversed twin swaps languages and the direction tag; (ii) every reversible instance has its twin in the same file, and every NEG instance is a singleton; (iii) a `pair_id` spans all tracks (2 rows in each monolingual track, 12 in T4). Each monolingual track has 3,042 training rows (1,282 reversible pairs + 478 NEG) and 660 dev rows (277 + 106); T4 has 18,252 / 3,960.

2 System

2.1 Cross-encoder

A pretrained encoder reads [CLS] a [SEP] b [SEP] and a linear head maps the first-token state to four logits. Backbones: DeBERTa-v3-base [2] (T1 only), mmBERT-base [3], mDeBERTa-v3-base fine-tuned on multilingual NLI, and XLM-R-large [4] fine-tuned on XNLI. Multilingual models are trained on all four tracks at once (27,378 rows), which already presents every source pair in nine language views.

2.2 Consistency-aware decoding

For instance x with log-probabilities $\ell(x) \in \mathbb{R}^4$, define its canonical score $\tilde{\ell}(x) = \ell(x)$ for **original** rows and $\tilde{\ell}(x) = P\ell(x)$ for **flipped** rows, where P swaps the FE and BE coordinates. Instances are grouped, the mean of $\tilde{\ell}$ over the group is decoded to one canonical label y^* (after adding a tunable bias β to the NEG score), and y^* is mapped back through P for flipped rows. Three groupings: *independent* (each instance alone, plain argmax), *twin* (instance + reversed twin), and *source* (all rows sharing a `pair_id` in the decoded file(s)). Any group containing both directions is decoded to a Rev-compatible pair by construction, so SoftCons is 1 on every pair not predicted as NEG. In all reported per-track numbers, *source* pools only the views present in that track’s own file; pooling across track files is discussed separately.

2.3 Symmetric (reversal-equivariant) architecture

As an architectural alternative we score both orders with the shared encoder f and output

$$g(a, b) = \frac{1}{2}(f(a, b) + P f(b, a)), \quad (1)$$

so $g(b, a) = P g(a, b)$ holds exactly for every input, at twice the compute (implemented as `symmetric: true`; a unit test checks equivariance to 10^{-5}).

2.4 Training objectives

Cross-entropy, optionally with label smoothing, plus a reversal-consistency term over twins, which a twin-aware batch sampler keeps in the same batch:

$$\mathcal{L} = \text{CE} + \lambda \text{KL}_{\text{sym}}(p(x) \| P p(x_{\text{rev}})). \quad (2)$$

Reverse-negative augmentation adds the reversed copy of every NEG pair (label-safe, since NEG relations are symmetric in this label set).

2.5 Structural prior

Fact (ii) makes the reversible/NEG decision recoverable from the file layout: a group with a reversed twin present is decoded among {EQ,FE,BE}, and a group without one is decoded as NEG. This is exact on train and dev. It is a property of how the data were built, not of the phrases, and could be judged an exploit; the test files may also be laid out differently. We therefore report every result with and without it. A *prior-aware* variant trains only on the three reversible labels, since the prior never asks the model about NEG.

3 Experimental setup

All runs use one NVIDIA RTX 4000 Ada (20 GB), bf16/fp16 mixed precision, AdamW, linear warm-up (10%) and decay, batch size 16–32, max length 64–128, and early stopping on the mean of the three official scores on dev. Because multilingual models peak within the first epoch, we validate

every 15% of an epoch (`epoch_fraction`). Learning rates: $2 \cdot 10^{-5}$ (DeBERTa-base), $5 \cdot 10^{-5}$ (mmBERT), $1.5 \cdot 10^{-5}$ (mDeBERTa, fine-grained), 10^{-5} (XLM-R-large). Symmetric XLM-R runs use batch size 8 to fit in memory. The training-time monitor (“dev score mean”, the mean of the three official scores) decodes the four dev files jointly, i.e. with cross-track pooling; it is used only for early stopping and for comparing runs, while all tables use per-track decoding. Ensembles average log-probabilities before decoding. Our metric implementation matches the official scorer to machine precision on every track and decoding mode.

4 Results

4.1 Decoding on a single model

Table 1 isolates the decoding effect on DeBERTa-v3-base (T1, one seed). Twin decoding trades 0.6 F1 for +5.4 SoftCons and +2.5 HardCons; SoftCons stays below 100 only because of NEG predictions on reversible pairs. With the structural prior the same model reaches exactly the leaderboard leader’s T1 scores (92.4 / 100 / 91.0), which suggests the leader uses the same prior.

Decoding (T1 dev, DeBERTa-v3-base)	W-F1	SoftCons	HardCons
independent	82.7	88.1	83.8
twin	82.1	93.5	86.3
source	82.1	93.5	86.3
twin + structural prior	92.4	100.0	91.0

Table 1: Decoding ablation on one model (official scorer confirms every row).

4.2 Backbones, architecture and training variants

Tables 2 and 3 give per-track dev scores (source decoding within each track) for single models and ensembles, without and with the structural prior.

Observations.

- **Backbone.** NLI-pretrained encoders (mDeBERTa-NLI, XLM-R-XNLI) clearly beat the MLM-only mmBERT-base, which overfits after one epoch; a Basque-only encoder (JaunBERT, T3 only) reached a dev score mean of only 76.9, far below the multilingual NLI models trained on all tracks.
- **Symmetric architecture.** Exact equivariance does not help a single model beyond what twin/source decoding already guarantees, and adding the symmetric mDeBERTa to the clean-7 ensemble slightly lowers it (80.8 / 95.0 / 87.3 vs. 80.8 / 95.7 / 88.0). Decoding-time consistency makes the architectural guarantee redundant here.
- **Reverse-negative augmentation hurts** on both backbones (Section 4.3). We have no verified explanation; one hypothesis is that the reversed copies double the weight of NEG examples and give them in-batch twins, which changes what the reversal-KL term sees.
- **English-only large model.** DeBERTa-v3-large (MNLI/FEVER/ANLI/ Ling/WANLI) trained on T1 reaches a dev score mean of 88.9, against 87.3 for DeBERTa-v3-base (83.3 / 95.3 / 88.1 on T1 with twin decoding).
- **Ensembles.** Averaging seeds and backbone families is the most reliable gain in both settings. Prior-aware members never learned NEG, so they are used only with the prior.

System	T1 EN			T2 ES			T3 EU			T4 Mixed			ALL-4		
	F1	S	H	F1	S	H	F1	S	H	F1	S	H	F1	S	H
mmBERT-base	77.4	87.7	78.3	74.1	91.7	78.0	67.8	88.8	73.6	72.5	83.4	74.4	72.9	87.9	76.1
mDeBERTa-v3-base NLI	83.6	95.3	89.2	79.3	94.2	86.6	77.4	93.5	84.1	78.9	94.2	85.9	79.8	94.3	86.5
mDeBERTa, fine-grained val. (A)	81.8	96.0	88.4	78.2	94.9	86.3	78.8	95.3	86.3	78.8	97.5	87.0	79.4	95.9	87.0
A + rev.-neg. aug.	83.4	92.1	86.3	81.1	92.4	84.5	76.4	89.2	79.1	80.0	91.3	82.7	80.2	91.2	83.1
A + NEG weight 2	82.8	92.4	85.9	80.5	93.9	85.9	76.7	88.4	79.4	78.7	90.3	81.2	79.7	91.2	83.1
A + focal ($\gamma=2$)	81.4	94.9	88.1	78.5	94.9	86.6	75.3	94.2	84.5	77.6	93.5	84.5	78.2	94.4	85.9
A + iSTS answers-students	80.6	96.0	88.4	76.8	95.3	85.6	76.2	95.7	85.9	77.5	97.8	88.1	77.8	96.2	87.0
A, symmetric arch.	82.1	93.1	85.6	78.6	93.5	84.1	77.2	92.4	82.7	79.4	94.2	84.8	79.3	93.3	84.3
XLM-R-large XNLI (B)	83.0	95.3	88.1	79.2	96.4	87.4	75.1	96.8	82.7	77.9	96.8	87.0	78.8	96.3	86.3
B + rev.-neg. aug.	78.5	92.1	83.0	75.1	93.9	80.9	70.1	91.0	75.5	73.3	93.9	79.1	74.3	92.7	79.6
B + iSTS answers-students	80.4	93.1	85.2	76.6	92.1	83.8	75.7	93.5	83.0	77.8	92.4	85.2	77.6	92.8	84.3
Ens. clean-7 (A, B + 5 seeds)	83.7	95.3	89.9	80.6	96.0	88.1	79.4	95.7	87.0	79.4	95.7	87.0	80.8	95.7	88.0

Table 2: Development results *without* the structural prior (F1 = Weighted F1, S = SoftCons, H = HardCons, in %). Single seeds unless marked as ensembles.

System	T1 EN			T2 ES			T3 EU			T4 Mixed			ALL-4		
	F1	S	H	F1	S	H	F1	S	H	F1	S	H	F1	S	H
mmBERT-base	87.1	100.0	84.8	85.2	100.0	82.7	82.1	100.0	79.8	88.5	100.0	86.6	85.7	100.0	83.5
mDeBERTa-v3-base NLI	92.3	100.0	91.0	90.5	100.0	88.8	89.6	100.0	87.7	90.9	100.0	89.2	90.8	100.0	89.2
mDeBERTa, fine-grained val. (A)	91.7	100.0	90.3	89.6	100.0	87.7	89.9	100.0	88.1	90.3	100.0	88.4	90.4	100.0	88.6
A + rev.-neg. aug.	92.6	100.0	91.3	90.9	100.0	89.2	87.0	100.0	84.5	90.9	100.0	89.2	90.4	100.0	88.5
A + NEG weight 2	92.1	100.0	90.6	90.6	100.0	88.8	87.9	100.0	85.6	90.0	100.0	88.1	90.2	100.0	88.3
A + focal ($\gamma=2$)	91.8	100.0	90.3	90.6	100.0	88.8	89.4	100.0	87.4	91.6	100.0	89.9	90.8	100.0	89.1
A + iSTS answers-students	92.1	100.0	90.6	89.6	100.0	87.7	89.4	100.0	87.4	90.6	100.0	88.8	90.4	100.0	88.6
A, symmetric arch.	91.2	100.0	89.5	89.6	100.0	87.7	88.2	100.0	85.9	90.7	100.0	88.8	89.9	100.0	88.0
XLM-R-large XNLI (B)	92.4	100.0	91.0	91.0	100.0	89.2	85.9	100.0	83.4	91.3	100.0	89.5	90.2	100.0	88.3
B + rev.-neg. aug.	91.0	100.0	89.2	88.3	100.0	85.9	85.3	100.0	82.3	86.6	100.0	83.8	87.8	100.0	85.3
B + iSTS answers-students	90.6	100.0	88.8	90.5	100.0	88.8	89.0	100.0	87.0	91.8	100.0	90.3	90.5	100.0	88.7
Ens. clean-7 (A, B + 5 seeds)	93.3	100.0	92.1	91.2	100.0	89.5	90.7	100.0	88.8	92.5	100.0	91.0	91.9	100.0	90.3
mDeBERTa, prior-aware	92.3	100.0	91.0	90.9	100.0	89.2	89.1	100.0	87.0	92.1	100.0	90.6	91.1	100.0	89.4
mDeBERTa, prior-aware, symmetric	93.0	100.0	91.7	91.8	100.0	90.3	90.1	100.0	88.1	92.5	100.0	91.0	91.9	100.0	90.3
XLM-R-large, prior-aware	91.8	100.0	90.3	91.0	100.0	89.2	88.5	100.0	86.3	91.9	100.0	90.3	90.8	100.0	89.0
XLM-R-large, prior-aware, symmetric	92.4	100.0	91.0	91.2	100.0	89.5	87.6	100.0	85.2	91.0	100.0	89.2	90.5	100.0	88.7
Ens. prior-aware-5	93.9	100.0	92.8	91.8	100.0	90.3	90.6	100.0	88.8	92.2	100.0	90.6	92.1	100.0	90.6
Ens. prior-aware-5 + clean-7	94.2	100.0	93.1	92.1	100.0	90.6	90.4	100.0	88.4	92.8	100.0	91.3	92.4	100.0	90.9
Ens. prior-14 (+ 2 symmetric)	94.2	100.0	93.1	92.4	100.0	91.0	91.3	100.0	89.5	92.5	100.0	91.0	92.6	100.0	91.2
Leaderboard #1 (dev)	92.4	100	91.0	92.8	100	91.3	90.3	100	88.4	90.2	100	88.2	91.4	100	89.7

Table 3: Development results *with* the structural prior, and the current CodaBench dev leader (29 Sep 2026).

- **T1.** The multilingual clean-7 ensemble already reaches 83.7 / 95.3 / 89.9 on T1. Adding the English-only DeBERTa-v3-large does not help (83.7 / 95.3 / 89.5), nor does it with the prior (prior-aware-5 + clean-7: 94.2 / 100 / 93.1).
- **Final systems.** *Clean:* clean-7, 80.8 / 95.7 / 88.0 ALL-4. *With prior:* prior-14 (5 prior-aware + 2 symmetric prior-aware + clean-7), 92.6 / 100 / 91.2 ALL-4; per track 94.2/93.1 (T1), 92.4/91.0 (T2), 91.3/89.5 (T3), 92.5/91.0 (T4) F1/HardCons. It leads the dev leaderboard on T1, T3, T4 and ALL-4, and trails it slightly on T2 (92.8/91.3). Both are packaged as CodaBench dev submissions.
- **neg is the bottleneck.** Without the prior, F1 stalls around 80 across all backbones, and nearly all of the gap to the prior-based systems is NEG vs. reversible confusion.

4.3 Targeting neg detection

Without the prior the three reversible labels are decoded well and almost all errors involve NEG. A literature scan (iSTS 2016 systems, HANS-style overlap heuristics, lexical-entailment failures, focal loss, intermediate-task transfer) suggested five cheap levers, which we tested against the same seed and configuration:

- All numbers below are ALL-4 F1 / SoftCons / HardCons with per-track decoding (Table 2); the references are mDeBERTa (A) 79.4 / 95.9 / 87.0 and XLM-R-large (B) 78.8 / 96.3 / 86.3.
- *Class-weighted CE* (NEG weight 2) on A: 79.7 / 91.2 / 83.1. *Focal loss* ($\gamma = 2$) on A: 78.2 / 94.4 / 85.9.
- *Reverse-negative augmentation*: A 80.2 / 91.2 / 83.1; B 74.3 / 92.7 / 79.6. On A it trades about 1 F1 point for 4–5 points of consistency; on B it is worse on every metric.
- *iSTS answers-students*: 1,010 chunk pairs converted to DiCo labels (EQUI→EQ, SPE1/2→FE/BE, SIMI/REL/OPPO→NEG; none overlap DiCo train/dev; the PhrasIS-source headlines/images subsets are excluded) added to training: A 77.8 / 96.2 / 87.0, B 77.6 / 92.8 / 84.3; a second seed of A was also lower on the training monitor (86.9 vs. 89.3).
- *Stacked NEG detector*: logistic regression / gradient boosting over the single-row log-probabilities of the clean-7 members plus lexical features (Jaccard overlap, containment, length, number mismatch, negation), fit with 5-fold cross-validation grouped by `pair_id`. Best: 81.4 / 91.8 / 84.7 (logistic, threshold 0.5) vs. 80.8 / 95.7 / 88.0 for the plain ensemble: slightly higher F1, clearly lower consistency.
- *NEG threshold*: Table 4.

The ensemble’s own NEG score already carries the lexical signal, and the remaining NEG errors look like genuinely hard “related but not entailing” pairs.

neg threshold. Table 4 sweeps the NEG bias β on the clean 7-model ensemble. Lowering β buys consistency with F1 and vice versa; the mean of the three scores is flat (88–89), so β is a knob for choosing a point on the trade-off, not a free gain.

β	F1	S	H
−1.5	76.9	99.3	90.3
−1.0	79.1	98.5	89.8
−0.5	80.2	97.5	89.2
0.0	80.8	95.7	88.0
+0.5	80.9	93.7	86.5
+1.0	80.7	91.5	84.5
+2.0	78.8	83.6	78.1

Table 4: ALL-4 dev scores of the clean-7 ensemble vs. NEG bias.

4.4 Cross-track pooling

A `pair_id` spans all four tracks, so decoding the four dev files jointly pools 18 views of each source pair. This helps further (e.g. 92.7 / 100 / 91.3 ALL-4 with the prior for a 4-model ensemble), but it uses other tracks’ inputs to predict a given track and makes all four tracks’ predictions identical up to language, so we exclude it from our headline numbers pending organizer guidance.

5 Discussion and limitations

- **Dev-set selection.** All model, ensemble and β choices were made on the dev set, which has only 277 reversible pairs per track (HardCons standard error ≈ 2.6 points). The reported numbers are therefore optimistic; grouped cross-validation over train+dev is the next step.
- **Legitimacy of the structural prior** and of cross-track pooling must be confirmed with the organizers before any official submission uses them. The clean system stands on its own.

- **Not yet explored:** adapters/LoRA for decoder LLMs, head initialisation from the 3-way NLI head, translate-test views, and an LLM teacher for NEG detection.

6 Reproducibility

Code: the project repository (configs in `configs/`, scripts in `scripts/`). Checkpoints, dev log-probabilities and run configs are in the private Hugging Face repository `LakoreAI/dico-nli-checkpoints`. All tables are regenerated from saved log-probabilities by `scripts/training/report_tables.py`.

References

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